Treatment of Hypovitaminosis D in Infants and Toddlers

Context: Hypovitaminosis D appears to be on the rise in young children, with implications for skeletal and overall health. Objective: The objective of the study was to compare the safety and efficacy of vitamin D2 daily, vitamin D2 weekly, and vitamin D3 daily, combined with supplemental calcium, in raising serum 25-hydroxyvitamin D [25(OH)D] and lowering PTH concentrations. Design: This was a 6-wk randomized controlled trial. Setting: The study was conducted at an urban pediatric clinic in Boston. Subjects: Forty otherwise healthy infants and toddlers with hypovitaminosis D [25(OH)D $<$ 20 ng/ml] participated in the study. Interventions: Participants were assigned to one of three regimens: 2,000 IU oral vitamin D2 daily, 50,000 IU vitamin D2 weekly, or 2,000 IU vitamin D3 daily. Each was also prescribed elemental calcium (50 mg/kg·d). Infants received treatment for 6 wk. Main Outcome Measures: Before and after treatment, serum measurements of 25(OH)D, PTH, calcium, and alkaline phosphatase were taken. Results: All treatments approximately tripled the 25(OH)D concentration. Preplanned comparisons were nonsignificant: daily vitamin D2 vs. weekly vitamin D2 (12% difference in effect, $P = 0.66$) and daily D2 vs. daily D3 (7%, $P = 0.82$). The mean serum calcium change was small and similar in the three groups. There was no significant difference in PTH suppression. Conclusions: Short-term vitamin D2 2,000 IU daily, vitamin D2 50,000 IU weekly, or vitamin D3 2,000 IU daily yield equivalent outcomes in the treatment of hypovitaminosis D among young children. Therefore, pediatric providers can individualize the treatment regimen for a given patient to ensure compliance, given that no difference in efficacy or safety was noted among these three common treatment regimens.